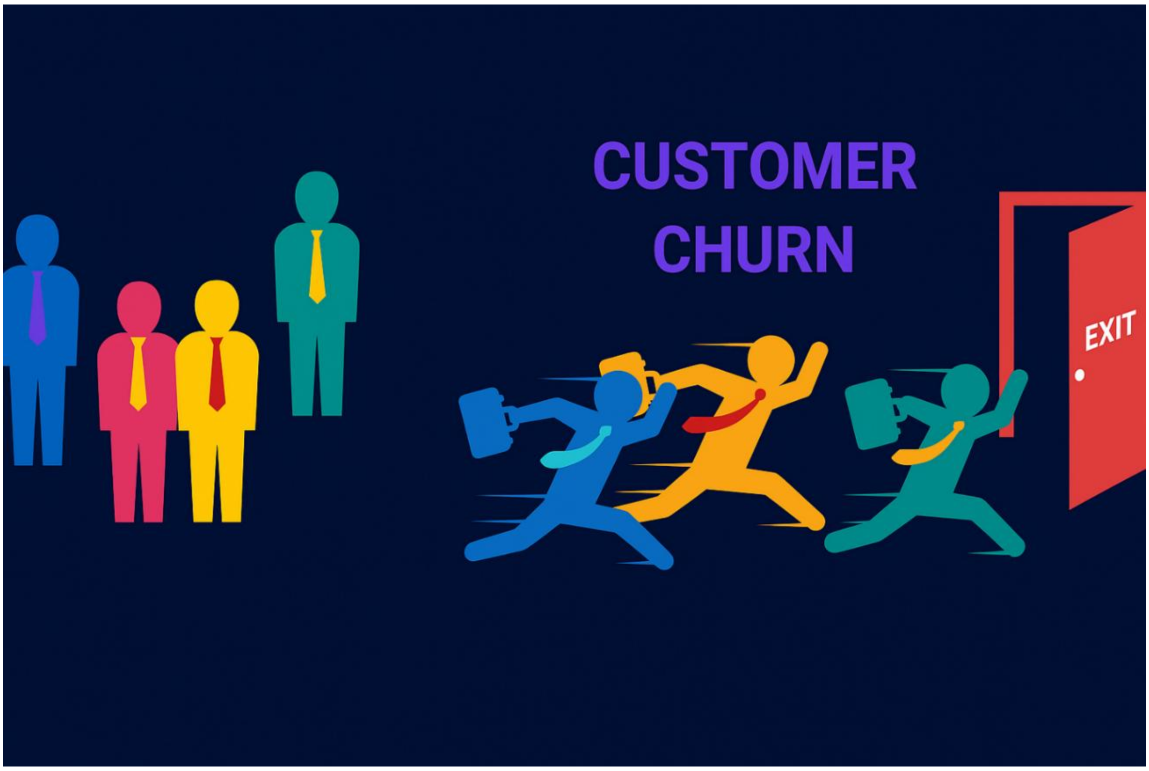
# Speedy Call Customer Churn Data Analysis.

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# Abstract.


- This study is based on customer churn patterns within SpeedyCall, a telecommunication company in USA, using a dataset of 7,043 customer records. The analysis focuses on identifying key demographic, behavioral, financial, and service-related factors that influence whether customers discontinue their services. Descriptive statistical techniques and visual analysis were used to examine relationships between customer characteristics and churn behavior.

- The results show that contract type and tenure are the most significant predictors of churn. Customers with month-to-month contracts and shorter tenure were much more likely to leave, while those with long-term contracts demonstrated higher loyalty. Senior citizens, and customers without partners or dependents, showed higher churn rates, indicating that age and household structure affect retention. Service-related factors, such as the absence of online security, backup, and technical support, were strongly linked to customer loss, whereas customers who subscribed to multiple services were more loyal. Financial analysis showed that churn was concentrated among customers with high monthly charges and low total charges, suggesting that new customers paying higher fees are more likely to leave early. The analysis also showed that customers who use electronic check payments have a significantly higher churn rate compared to other payment methods, indicating that billing method is an important factor in customer retention.

- Overall, the study highlights the importance of contract design, service bundling, pricing strategy, and customer involvement in reducing churn and improving retention. These insights can guide SpeedyCall in developing targeted retention programs to enhance long-term profitability and customer satisfaction.

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# 1. Introduction.

- In the modern telecommunication industry, customer retention has become a crucial factor in maintaining profitability and competitiveness. Customer churn refers to the situation where customer discontinue services from a specific company. This is because there are numerous service providers offering similar packages and there is competitive pricing, service quality issues and customer preferences. These challenges make it difficult for companies to maintain a stable customer base.

- For a company, new customers typically cost more than retaining existing ones. Therefore, understanding the current customer base and identifying the reasons behind churn are essential for long-term success. SpeedyCall is a telecommunication company in USA which aims to analyze its customer base to identify factors influencing churn. In addition to reducing revenue loss, churn prediction improves customer satisfaction through continuous involvement and better service strategies.

# 2. Objectives.

- To determine the demographic, behavioral, and service-related differences between churners and non-churners.

- To assess the financial differences between churned and non-churned customers by comparing MonthlyCharges and TotalCharges.

- To evaluate the relationship between service subscriptions and customer retention.

# 3. Data.

- The dataset “SpeedyCall Customer Churn Data.xlsx” was used for this analysis. It contains customer details of a telecommunication company named SpeedyCall in USA. In this data set there are 7043 observations. The dataset consists of both categorical and numerical variables that describe customer behavior and their relationship with the company. This contains 21 variables, which are 3 quantitative variables and 17 qualitative variables to describe the observations. Tenure, Monthly Charges and Total Charges are the three quantitative variables. All the other variables are qualitative variables.

| Variable Name | Description |
| --- | --- |
| **customerID** | Customer ID |
| **gender** | Whether the customer is a male or a female |
| **SeniorCitizen** | Whether the customer is a senior citizen or not (1, 0) |
| **Partner** | Whether the customer has a partner or not (Yes, No) |
| **Dependents** | Whether the customer has dependents or not (Yes, No) |
| **tenure** | Number of months the customer has stayed with the company. |
| **PhoneService** | Whether the customer has a phone service or not (Yes, No) |
| **MultipleLines** | Whether the customer has multiple lines or not (Yes, No, No phone service) |
| **InternetService** | Customer’s internet service provider (DSL, Fiber optic, No) |
| **OnlineSecurity** | Whether the customer has online security or not (Yes, No, No internet service) |
| **OnlineBackup** | Whether the customer has online backup or not (Yes, No, No internet service) |
| **DeviceProtection** | Whether the customer has device protection or not (Yes, No, No internet service) |
| **TechSupport** | Whether the customer has tech support or not (Yes, No, No internet service) |
| **StreamingTV** | Whether the customer has streaming TV or not (Yes, No, No internet service) |
| **StreamingMovies** | Whether the customer has streaming movies or not (Yes, No, No internet service) |
| **Contract** | The contract term of the customer (Month-to-month, One year, Two year) |
| **PaperlessBilling** | Whether the customer has paperless billing or not (Yes, No) |
| **PaymentMethod** | The customer’s payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic)) |
| **MonthlyCharges** | The amount charged to the customer monthly |
| **TotalCharges** | The total amount charged to the customer |
| **Churn** | Whether the customer churned or not (Yes or No) |

*Figure 1: Table of Variables.*

# 4. Data Analysis.

- In this section, a descriptive analysis was conducted to draw conclusions about how influential factors affect customer churn. There were eleven missing values in the dataset. Before performing the analysis, the missing values were removed.

- To determine the demographic, behavioral, and service-related differences between churners and non-churners.

<u>DEMOGRAPHIC VARIABLES.</u>

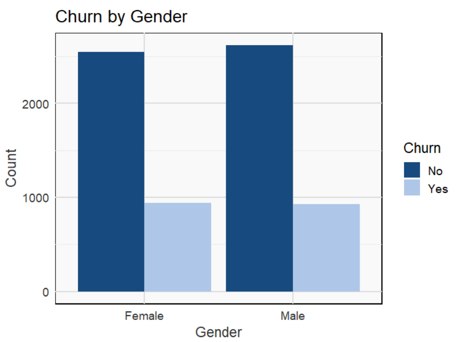


*Figure 2: The Plot of Churn by Gender.*

This bar chart shows the count of churned (Yes) and non-churned (No) customers, segmented by gender. Total number of non-churners is substantially larger than that of churners. Therefore, chart shows that gender is not a predictor of churn, as males and females churn at almost identical rates.

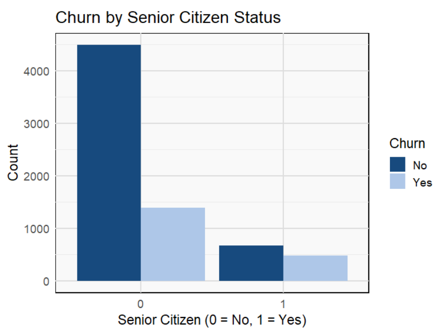


*Figure 3: The Plot of Churn by Senior Citizen*

This bar graph clearly shows that senior citizens churn at a significantly higher rate than younger customers. While the total number of senior citizens (1) is much smaller than non-seniors (0), the proportion of churners within their bar is far larger. This suggests that senior citizenship is a significant factor in determining whether a customer will leave.

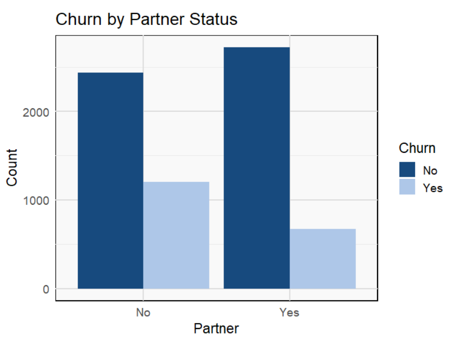


*Figure 4: The Plot of Churn by Partner Status.*

This chart shows that customers without a partner churn at a noticeably higher rate than customers who have one. The proportion of churners is visibly larger for the "No" group, suggesting that having a partner is associated with higher customer retention.

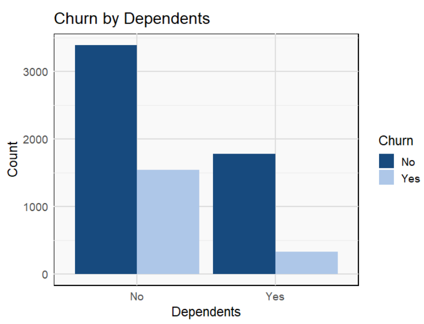


*Figure 5: The Plot of Churn by Dependents.*

This chart shows that customers without dependents are significantly more likely to churn. The proportion of customers with dependents is visibly smaller within the Churn - Yes bar compared to their proportion in the Churn - No bar. This implies that having dependents is associated with higher customer loyalty.

## BEHAVIORAL VARIABLES.

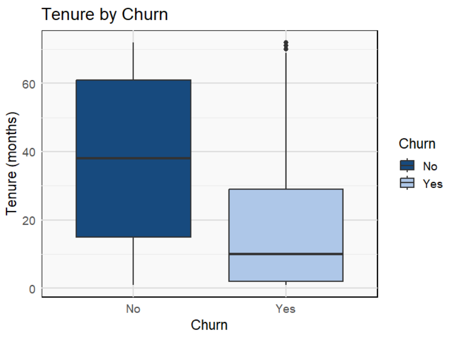


*Figure 6: The Plot of Churn by Tenure.*

This chart shows that new customers have the highest churn rate. As the number of months, a customer stays with the company (tenure) increases, their loyalty significantly improves, and the likelihood of them churning drops dramatically.

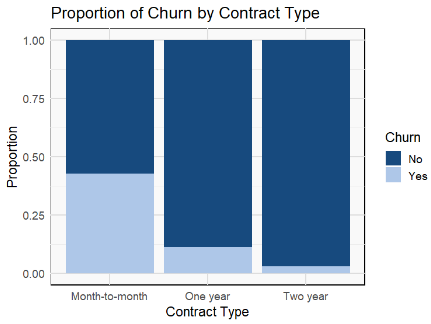


*Figure 7: The Plot of Churn by Contract Type.*

This chart clearly shows that customer tenure is a powerful predictor of loyalty. The churn rate is highest among new customers and decreases significantly as the customer's tenure increases. This indicates that long-term customers are far less likely to churn.

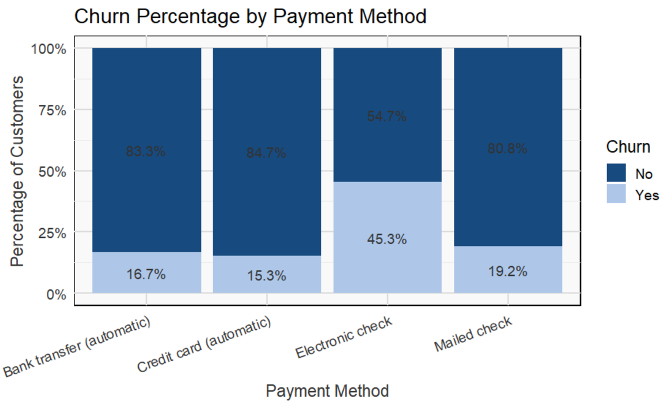


*Figure 8: The Plot of Churn Percentage by Payment Method.*

This chart shows that customers paying by Electronic check have a dramatically higher churn rate. In contrast, customers using Bank transfer and Credit card are the most loyal and have the lowest churn rates.

## SERVICE-RELATED VARIABLES.

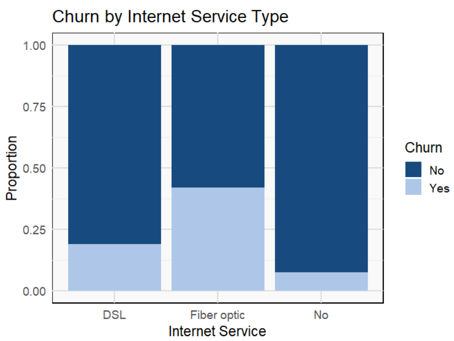


*Figure 9: The Plot of Churn by Internet Service Type.*

This chart shows that customers with Fiber optic internet have a dramatically higher churn rate. Compared to DSL or not having internet service, the Fiber optic category is proportionally much larger in the Churn - Yes bar, indicates that it is a significant driver of customer loss.

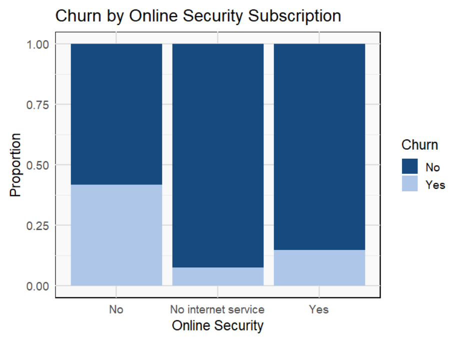


*Figure 10: The Plot of Churn by Online Security Subscription.*

This chart shows that customers who do not subscribe to OnlineSecurity churn at a significantly higher rate. This indicates that not having this add-on service is a key risk factor for customer loss.

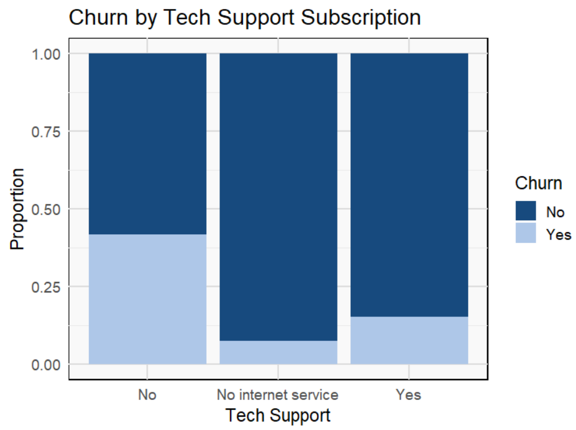


*Figure 11: The Plot of Churn by Tech Support Subscription.*

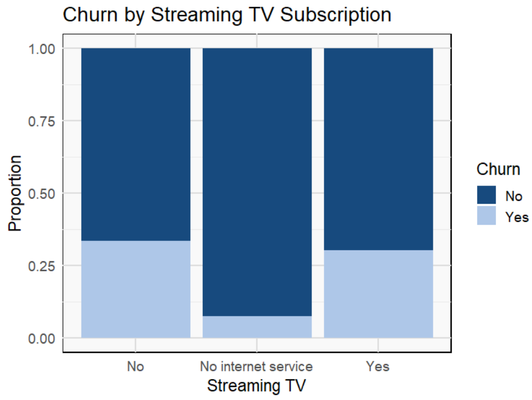


*Figure 12: The Plot of Churn by Streaming TV Subscription.*

This chart clearly shows that customers without TechSupport churn at a significantly higher rate, making it a key risk factor.

This chart shows that subscribing to StreamingTV is associated with higher customer retention. This indicates that customers who have internet but choose not to get the StreamingTV add-on are at a slightly higher risk of churning.

- To assess the financial differences between churned and non-churned customers by comparing MonthlyCharges and TotalCharges.

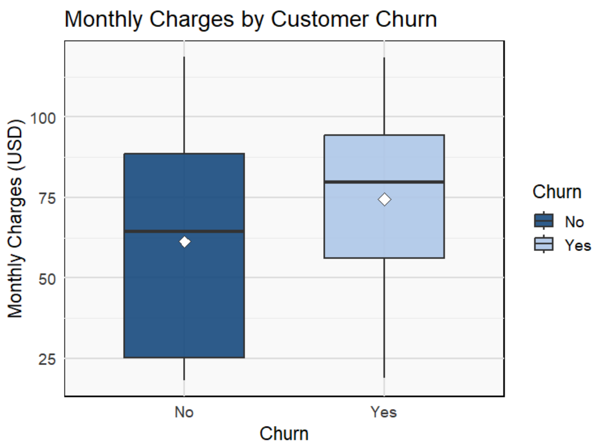


*Figure 13: The Plot of Monthly Charges by customer churn.*

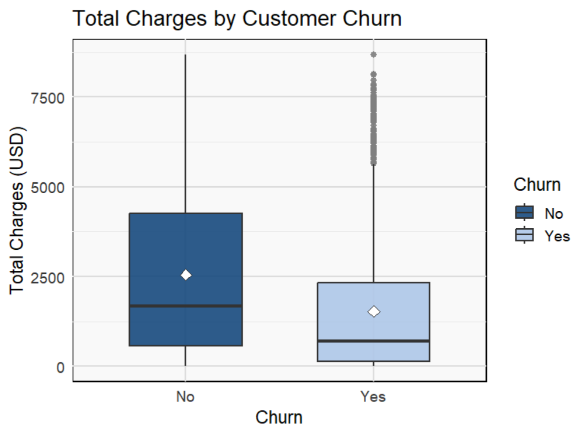


*Figure 14: The Plot of Total Charges by Customer Churn.*

This chart shows that customers with high monthly charges have the highest churn rate. In contrast, customers with the lowest monthly charges are the most loyal and very unlikely to churn. This indicates that high price is a major driver of customer loss.

This chart shows that churn is heavily concentrated among customers with the lowest TotalCharges. It indicates that new customers are the most likely to churn because they have not had time to accumulate high total charges.

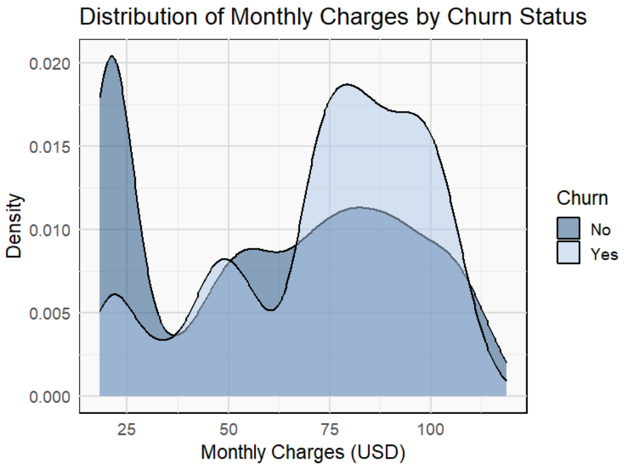


*Figure 15: Distribution of Monthly Charges by Churn Status.*

This plot shows that customers with higher MonthlyCharges are much more likely to churn. The Churn - Yes is significantly higher than the Churn - No curve in the $70-$105 range. In contrast, customers with low monthly charges are very loyal, as shown by the high dark blue peak on the left.

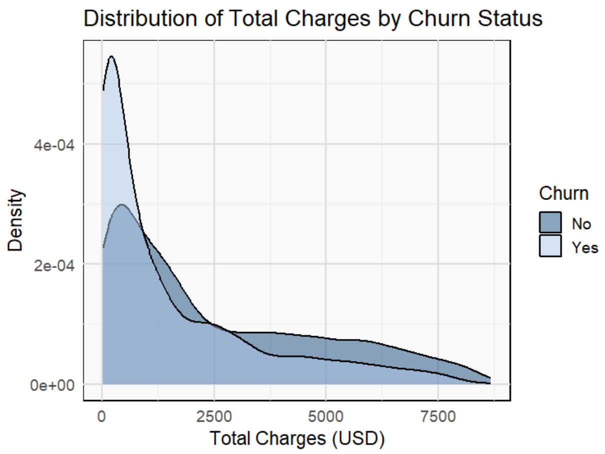


*Figure 16: Distribution of Total Charges by Churn Status.*

This chart clearly shows that customers who churn are almost all new customers with very low TotalCharges. This indicates that customers are leaving very early in their service, before they have accumulated significant billing.

- To evaluate the relationship between service subscriptions and customer retention.

Service relates columns were selected and a new variable was created as number of subscribed services. It was created as convert ‘Yes’/ ‘No’ to 1/0, then sum across services for each customer.

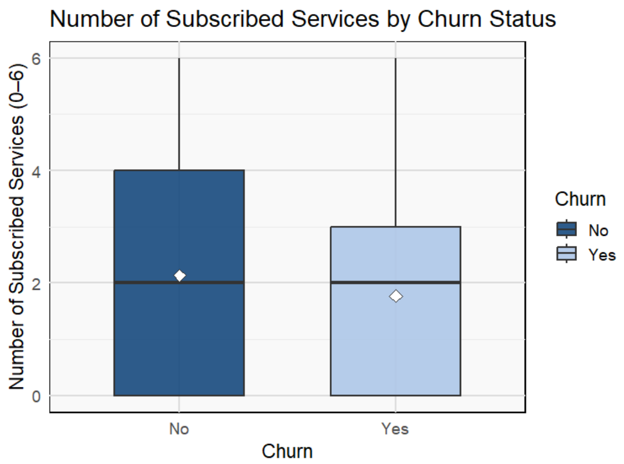


*Figure 17: Number of Subscribed Services by Churn Status.*

This boxplot shows that customers who churned (Yes) tend to have fewer subscribed services. The median number of services for churners is visibly lower (around 2 services) than for loyal customers (around 3). This suggests that customers who are more integrated into the service ecosystem with more add-ons are less likely to leave.

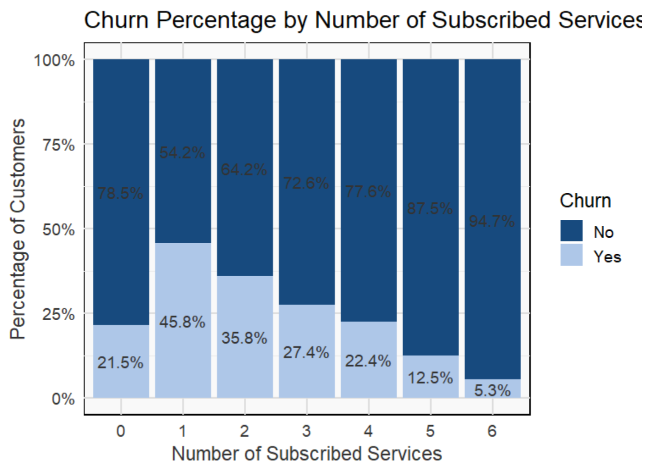


*Figure 18: Churn Percentage od Number of Subscribed Services.*

This chart clearly shows that customers with only one service have the highest churn rate by far, at 45.8%. Customers with two services (35.8% churn) come next. The key finding is that customers' churn rate significantly decreases as they sign up for more services (from two to six), implies that customers who are more integrated into the service ecosystem are considerably more loyal.

# 5. General Discussion and Conclusion.

- According to the analysis contract type and tenure are the strongest predictors of customer churn. Customers with month-to-month contracts had the highest likelihood of leaving the company, while those with one-year or two-year contracts showed greater loyalty. Similarly, longer tenure was strongly associated with lower churn, emphasizing that customer retention improves significantly over time.
- Demographic factors also played an important role. Senior citizens exhibited a higher churn rate compared to younger customers, which could be due to challenges in adapting to modern digital services or increased sensitivity to service costs. Customers without partners or dependents were more likely to churn, indicates that individuals with family responsibilities tend to maintain longer and more stable service relationships. Additionally, the payment method analysis revealed that electronic check users churn at the highest rate among all payment categories, implies that payment convenience and billing preferences may influence customer stability.
- In terms of service-related factors, the lack of online security, online backup, and technical support subscriptions was strongly linked to higher churn rates. On the other hand, customers who subscribed to multiple additional services showed a much lower tendency to leave. This suggests that service integration, bundling, and cross-selling are useful strategies for improving client retention.

- From a financial perspective, churn was most common among customers with high monthly charges and low total charges. This pattern suggests that many new customers paying higher fees leave early in their contracts, likely due to dissatisfaction with the perceived value or cost of the service. Meanwhile, customers with lower monthly charges and accumulated total charges demonstrated higher loyalty.
- In general, the results show that customer churn is influenced by a combination of contract commitment, service engagement, financial satisfaction, and demographic characteristics. SpeedyCall can reduce churn by encouraging long-term contracts, offering value-driven bundled packages, and improving service quality and affordability for weaker customer segments such as seniors and new users.